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Comparative Study of Wear Resistance and Surface Roughness of Injectable Versus Conventional Composite Resin - In Vitro Study

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DECLARATION

I declare that this thesis is my own work. It has not been submitted, in whole or in part, for any other degree or qualification. The experimental data are those of the laboratory study described herein. Sources used in the text are cited in the list of references.

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DEDICATION

To my family.



ABSTRACT

Background. Highly filled injectable composites are sold for load-bearing use. Their loss of mass and change in surface roughness after toothbrush abrasion still need a direct comparison with a conventional nanohybrid resin under the same laboratory conditions.

Aim. To compare percentage weight loss and arithmetic mean roughness (Ra) of Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS after 10 000 toothbrushing cycles.

Methods. Thirty-six discs (10 mm diameter $\times$ 1 mm thick; $n = 12$; shade A2) were packed in a CAD/CAM Teflon mould, cured through a Mylar strip and stored in water at 37 °C for 24 h. They were brushed at 2 N for 10 000 cycles in a Colgate Total slurry (RDA 70; 250 g/L). Mass was recorded on a RADWAG AS 220-R2. Ra was the mean of five MarSurf PS10 traces. Percentage weight loss was compared by one-way ANOVA. Ra after brushing was compared by the Kruskal–Wallis test ($\alpha = 0.05$).

Results. Mean percentage weight loss was 1.49 ± 3.14 % for Beautifil Flow Plus X F00, 0.65 ± 0.37 % for G-aenial Universal Injectable and 0.41 ± 0.25 % for Beautifil II LS ($p = 0.329$). Mean Ra after brushing was 0.104 ± 0.017 μm , 0.100 ± 0.023 μm and 0.194 ± 0.050 μm respectively ($p < 0.001$). Each injectable resin was smoother than Beautifil II LS. The two injectables did not differ from each other. One Beautifil Flow Plus X F00 disc lost about 11.4 % of its mass and accounts for that group's large standard deviation.



Conclusions. Under this toothbrushing protocol, gravimetric wear did not differ significantly among the three resins. Both injectable materials remained significantly smoother than the conventional nanohybrid. The injectable means stayed below the 0.2 μm plaque threshold. The nanohybrid mean sat on that line.

Keywords. Injectable composite; nanohybrid composite; toothbrush abrasion; wear; surface roughness; giomer.



الملخص

الخلفية. تُسوّق الراتينجات المركبة القابلة للحقن عالية الملاء للاستخدام تحت الحمل. ما زال فقد الكتلة وتغيّر خشونة السطح بعد سحل فرشاة الأسنان يحتاجان إلى مقارنة مباشرة مع راتينج نانوهجين تقليدي تحت الشروط المختبرية نفسها.

الهدف. مقارنة النسبة المئوية لفقد الوزن وخشونة السطح المتوسطة (Ra) لكل من Beautifil G-aenial Universal Injectable و Flow Plus X F00 و Beautifil II LS بعد عشرة آلاف دورة تفريش.

الطريقة. أُعد ستة وثلاثون قرصاً (قطر 10 مم × سمك 1 مم؛ ن = 12؛ الدرجة A2) في قالب تفلون، ثم فُرشت بحمولة 2 نيوتن في معلق Colgate Total وفق إرشاد ISO. سُجّلت الكتلة قبل السحل وبعده. قيس Ra بجهاز تماسي (خمس قراءات لكل قرص). قورن فقد الوزن بتحليل التباين الأحادي، وقورن Ra بعد التفريش باختبار كروسكال-واليس ($\alpha = 0.05$).

النتائج. كان متوسط فقد الوزن $1.49 \pm 3.14\%$ لـ Beautifil Flow Plus X F00، و $0.65 \pm 0.37\%$ لـ G-aenial Universal Injectable، و $0.41 \pm 0.25\%$ لـ Beautifil II LS ($p = 0.329$). وكان متوسط Ra بعد التفريش 0.104 ± 0.017 ميكرومتر، و 0.100 ± 0.023 ميكرومتر، و 0.194 ± 0.050 ميكرومتر على الترتيب ($p < 0.001$). كان كل راتينج قابل للحقن أنعم من Beautifil II LS، ولم



يختلف القابلان للحقن أحدهما عن الآخر. فقد قرص واحد من Beautifil Flow Plus X F00 نحو 11.4%

من كتلته وهو مصدر الانحراف المعياري الكبير في تلك المجموعة.

الاستنتاجات. تحت بروتوكول التفريش هذا لم يختلف السحل الوزني بين الراتينجات الثلاثة

اختلافاً يعتد به إحصائياً. بقي القابلان للحقن أنعم من النانو هجين التقليدي. بقيت متوسطات القابلين للحقن

دون عتبة اللويحة 0.2 ميكرومتر. وقع متوسط النانو هجين على تلك العتبة.

الكلمات المفتاحية. الراتينج المركب القابل للحقن؛ الراتينج النانو هجين؛ سحل فرشاة الأسنان؛

التآكل؛ خشونة السطح؛ الجايومر.



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LIST OF ABBREVIATIONS

ANOVA	Analysis of variance
CAD/CAM	Computer-aided design / computer-aided manufacture
ISO	International Organization for Standardization
LED	Light-emitting diode
Ra	Arithmetic mean roughness
RDA	Relative dentine abrasivity
SD	Standard deviation
S-PRG	Surface pre-reacted glass-ionomer
ΔRa	Change in Ra (after – before)



CHAPTER 1

INTRODUCTION

Resin composites are the usual material for direct restorations because they can be matched to tooth colour and bonded to enamel and dentine (11,27). They still fail at the surface. Wear flattens anatomy and can shorten the vertical dimension. Roughness holds plaque, takes stain and favours caries at the margin (1,2,17). Those two surface changes run together. The resin-rich skin is removed first. Filler then stands proud or is pulled out. Mass falls. Ra rises (3,4,28).

Toothbrushing is a daily source of both. Dentifrice abrasive works as a third body between the bristle and the restoration (3,8,29). On free surfaces of anterior restorations that is the main wear mode. Two-body contact wear is a different problem. It needs an antagonist and a much higher load. The two tests answer different questions and cannot be substituted for each other (3,6,10,34).

First-generation flowable resins were not written for load-bearing work. Filler load sat at about 40–55 wt%. Shrinkage was higher and wear was faster than in a packable hybrid (5,13). Highly filled injectable resins were made so that a syringeable paste could be used in cavities that used to be reserved for those packable composites. Filler load is raised, particle size is reduced, and silane treatment is improved until the paste still flows but no longer behaves like a classical low-filled flowable (5,6,20). Manufacturers now sell them for posterior as well as anterior use.

G-aenial Universal Injectable (GC Corporation, Tokyo, Japan) and Beautifil Flow Plus X F00 (Shofu Inc., Kyoto, Japan) are two such pastes. Beautifil II LS (Shofu Inc.) is a sculptable low-shrinkage nanohybrid giomer and is the conventional control in this work.



All three were used in shade A2. The approved protocol listed their filler volume as 50 %, 47 % and 68 % respectively.

Two of the three pastes are giomers. Beautifil Flow Plus X F00 and Beautifil II LS contain surface pre-reacted glass-ionomer (S-PRG) filler. That glass can release fluoride and other ions, and the same ion exchange can roughen the surface in acid (19,21). The present test does not measure ion release. G-aenial Universal Injectable is not a giomer. It is a highly filled dimethacrylate paste with ultrafine barium glass and silica (6,18). The comparison is therefore one injectable giomer and one injectable nanofill against a packable giomer nanohybrid, not three products of the same chemistry.

Independent data on these three products, tested together, under three-body toothbrush abrasion, are thin. Two-body chewing-simulator studies are not a substitute. They load a steatite antagonist at about 49 N. They measure volume, not toothbrush slurry (6). Some highly filled flowables have been reported as weaker than a paste; others have not (6,20). The present test is the other wear mode: ISO toothbrushing guidance, 2 N, 10 000 strokes, taken as about one year of twice-daily brushing (3,7,8).

An Ra of 0.2 μm is the usual threshold above which plaque retention rises (1,17). Surfaces smoother than that do not keep less plaque in a way that matters clinically. A restoration rougher than about 0.5 μm can be felt by the tongue (2). Those two numbers are the clinical frame for the roughness results.

Wear resistance of a composite is governed by filler volume, size, shape and coupling, by the resin and by the degree of conversion (11,12,26). Hydrolytic softening of the matrix can add to that loss after water storage (25,33). A laboratory ranking after a defined slurry and cycle count is useful. It is not a clinical lifetime (10,34). The question



here is narrower: after one conventional laboratory year of toothbrushing, do the two injectables lose more mass, or finish rougher, than Beautifil II LS?

1.1 Aim of the Study

To compare wear resistance, as percentage weight loss, and surface roughness (Ra) of two highly filled injectable composites with a conventional nanohybrid composite after standardised toothbrushing.

1.2 Objectives

1. To measure specimen mass before and after 10 000 brushing cycles and to calculate percentage and absolute weight loss.
2. To measure Ra before and after the same abrasion and to calculate ΔRa .
3. To compare the three materials with the statistical tests appropriate to each outcome.

1.3 Null Hypotheses

1. There is no significant difference in percentage weight loss among Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS.
2. There is no significant difference in Ra after toothbrushing among the same three materials.



CHAPTER 2

REVIEW OF LITERATURE

2.1 Wear of Restorative Resins

Wear is progressive loss of substance from two surfaces in relative motion (9). In the mouth the process is mixed. Attrition is two-body contact at sites that meet. Abrasion at non-contact sites is usually three-body, with food or toothpaste as the loose agent. Corrosion and fatigue run beside both (9,10). Laboratory ranking is useful. It is not a clinical lifetime (10,34).

Filler volume, size, shape, inter-particle spacing, the resin, the degree of conversion and the silane layer all matter (11,12,26). Condon and Ferracane varied cure, filler level and silane treatment in an oral-wear simulator. Wear rose once filler volume fell below 48 vol%. It rose in a straight line as the fraction of silane-treated filler was cut. Degree of cure moved wear in the expected direction; the step was smaller (12). Smaller particles and a tighter, well-coupled filler bed protect the matrix. Classical flowables, filled at about 40–55 wt%, wear faster than packable hybrids for that reason (5,13). Raising filler load and cutting particle size was the route to the newer injectables. G-aenial Universal Injectable is listed at 50 vol% in the approved protocol. That sits just above Condon's 48 vol% step. Beautifil II LS is listed at 68 vol% in the same table.

Water storage softens the resin and can loosen the silane layer. Ferracane reviewed hygroscopic and hydrolytic effects in dental polymer networks (25). Söderholm and co-workers showed hydrolytic degradation of the filler–matrix interface in earlier composites (33). A 24 h soak at 37 °C, as used here, is a short water challenge. It is not a long ageing study.



Toothbrushing tests follow ISO/TR 14569-1. The brush load should lie between 0.5 N and 2.5 N. The slurry is made from a dentifrice that meets ISO 11609 (3,14). Ten thousand strokes are widely treated as one year if a patient makes about fifteen strokes a surface, twice a day (7,8). The conversion is an estimate. It is still the figure most toothbrushing papers use. Heintze later set out how a wear simulator should be qualified if the ranking is to be compared across laboratories (34).

Wear may be read as volume, profile depth or mass. Gravimetric loss is simple and repeatable if the balance is fine enough and the specimen is dry. It does not describe the shape of the worn surface. Volume from a scan does. Heintze compared mass, profile depth and scanned volume and showed that the three methods do not tell the same story (15). Lambrechts reviewed the range of laboratory wear machines and warned against reading any one of them as a clinical year (10). The two numbers answer different questions.

2.2 Surface Roughness

Ra is the arithmetic mean of the absolute deviations of the profile from the centre line. It is the parameter defined for the profile method in ISO 4287 and the one almost every dental paper reports (16). Areal parameters are set out in ISO 25178-2 (23). They need an optical or scanning map. They are not required to rank Ra after toothbrushing.

Contact stylus profilometry remains a standard method. Kakaboura and co-workers compared profilometry, scanning electron microscopy, atomic force microscopy and gloss on the same composite surfaces. The methods agreed on rank more often than on the absolute number (31). Optical methods add a map. They are not required to rank Ra (4,16,31).



Bollen, Lambrechts and Quirynen set 0.2 μm as the plaque threshold (1). Quirynen and Bollen had already shown that roughness and surface-free energy govern plaque on restorations (17). Jones, Billington and Pearson later asked volunteers to rank composite discs with the tongue. Discrimination sat between 0.25 μm and 0.50 μm . They advised finishing to no more than 0.50 μm if the patient is not to feel the surface (2).

Toothbrushing raises Ra. Heintze and co-workers showed that the rise depends on material, time and load (4). Turssi and colleagues tracked resin-based materials during brushing and recorded a progressive rise in roughness (30). Nanofilled resins, with small particles that wear with the matrix, usually stay smoother than nanohybrids, in which larger fillers are uncovered or lost (8,18,28,35). The toothpaste matters. A higher relative dentine abrasivity cuts gloss faster and leaves a rougher track (8,29). da Costa and Ferracane's group showed that dentifrice type changes both Ra and gloss of the same composite (29). Monteiro and Spohr reached the same conclusion with a different set of pastes (8).

2.3 Filler Size and Toothbrush Abrasion

Cavalcante, Masouras, Watts, Pimenta and Silikas brushed model composites whose fillers ran from 100 nm to 1000 nm. After 20 000 strokes, larger fillers left a higher Ra and a lower gloss. Colour change was small and was not tied to filler size (28). That is the geometry argument in a single experiment.

Moraes and co-workers brushed packable, microhybrid, nanohybrid and microfilled resins. Wear and roughness were material-dependent. The microfilled paste behaved differently from the hybrids (32). de Oliveira and colleagues later compared nanofilled and microhybrid resins after 100 000 strokes. Nanofilled materials generally wore less and



stayed smoother, though not every product followed the mean (36). Senawongse and Pongprueksa polished and then brushed nanofill and nanohybrid pastes. After brushing the nanofill stayed smoother (35). Suzuki and Finger's group used a calcium-carbonate slurry for 50 000 cycles. Wear depth rose almost in a straight line with cycle count. Surface texture on SEM showed uniform abrasion of a nanofill and matrix loss around larger glass in two nanohybrids (37).

Taken together, those papers support a simple rule. When the filler is small and well coupled, the surface wears evenly. When larger glass or pre-polymerised particles sit in a softer matrix, the matrix goes first and Ra rises (4,28,35,37).

2.4 Injectable Composite Resins

Early flowables were not intended for occlusal load (5,13). Baroudi and Rodrigues reviewed that generation and kept them for lining, small repairs and conservative boxes (13). The current injectables are a different paste. G-ænial Universal Injectable is filled to about 69 wt% (50 vol%) with ultrafine barium glass and silica, mean size about 150 nm, in a UDMA / Bis-MEPP / TEGDMA matrix (6,18). Beautifil Flow Plus X F00 is a zero-flow giomer injectable with multifunctional glass and nano S-PRG filler (19). The protocol lists 47 vol%. It is not the same product as Beautifil Flow Plus F00 used in several earlier papers (6).

Ilie and Hickel surveyed mechanical behaviour across commercial composites and showed wide scatter inside each marketing class (26,27). A label such as “injectable” or “nanohybrid” does not fix wear or Ra. The filler bed does.

Rajabi et al. compared G-ænial Universal Injectable and Beautifil Flow Plus F00 with a conventional flowable and a nanohybrid paste in two-body wear (200 000 cycles, 49



N, steatite). The two injectables lost less volume than the paste and the classical flowable. Flexural strength was also higher. The two injectables did not differ from each other (6). That is occlusal contact wear, dry, measured as volume. It cannot be read as toothbrush mass loss.

Hasan, Eltoukhy and Zaghloul brushed G-ænial Universal Injectable, a nanofilled paste and a nanohybrid. After times taken as six months and one year, the injectable had the lowest Ra. The nanohybrid was the roughest. The two nanofilled resins did not differ (18). That design is closer to the present study.

Tzimas and co-workers reviewed highly filled flowables used as sole restoratives. Ra values were often statistically comparable to nanohybrid pastes, and sometimes lower, depending on filler packing and polish (20). Material-to-material scatter remains the rule.

2.5 Conventional Nanohybrid Resins and Giomers

Nanohybrid pastes mix nano-sized particles with larger submicron or micron fillers. Total inorganic load commonly exceeds 70–80 wt% or, in volume terms, sits well above the 48 vol% step that Condon described (11,12). Beautifil II LS is such a paste. The approved protocol lists 68 vol%. The resin uses Bis-GMA, UDMA, Bis-EMA, PEGMA and TEGDMA. The filler includes multifunctional glass and S-PRG (19). S-PRG releases ions, including fluoride. The same ion exchange can roughen the surface in acid (21). Larger and irregular fillers, once the matrix is brushed off, leave a higher Ra than a uniform nano-filled injectable (18,28,35).

An eight-year clinical series of a giomer paste (Beautifil with a self-etch primer) showed acceptable retention and little secondary caries in the teeth that were still available for recall (22). That is clinical behaviour of an earlier giomer, not toothbrush Ra of



Beautifil II LS. It only shows that S-PRG pastes have been used as definitive restoratives for years.

Packable nanohybrids remain the clinical reference for posterior direct work (11,27). They are harder to adapt in a narrow box and more likely to trap voids. The injectable products were written to close that handling gap without giving up wear resistance. Whether they do so under toothbrush abrasion is the question of this thesis.

ISO 4049 sets requirements for polymer-based restorative materials. It is the product standard, not a roughness standard (24). Roughness parameters remain those of ISO 4287 and, for maps, ISO 25178-2 (16,23).

2.6 Statement of the Problem

Two-body wear of G-ænial Universal Injectable and Beautifil Flow Plus F00 is favourable in one recent laboratory study (6). Toothbrush Ra of G-ænial Universal Injectable is favourable against a nanohybrid in another (18). Beautifil Flow Plus X F00 and Beautifil II LS have not been placed beside G-ænial Universal Injectable in the same 10 000-cycle, 2 N, gravimetric-plus-Ra protocol. That is the gap the experiment fills.



CHAPTER 3

MATERIALS AND METHODS

3.1 Study Design and Setting

The study was an in-vitro comparison of three resin composites. No patient, extracted tooth or identifiable tissue was used. Disc preparation was done in the laboratory of the Faculty of Dentistry, Pharos University in Alexandria. The toothbrushing run and the measurements were done in the Conservative Dentistry laboratory of the Faculty of Dentistry, Alexandria University.

The design follows the approved protocol of 3 February 2025. Wear is three-body toothbrush abrasion after ISO/TR 14569-1, not two-body antagonist wear (3,34). The materials fall under ISO 4049 as polymer-based restoratives (24). Roughness is reported as Ra after ISO 4287 (16).

3.2 Sample Size

Thirty-six discs were prepared, 10 mm in diameter and 1 mm thick. The discs were allocated at random to three groups of twelve. Disc geometry is that of the approved protocol. It is a flat face for profilometry and a known volume for weighing. It is not a Class II box.

The first sample-size estimate used G*Power version 3.1.9.2. It was based on a two-body wear comparison of highly filled injectables with a conventional paste (6). With 80 % power and $\alpha = 0.05$ the calculation gave ten discs a group (thirty in all). Twelve were used. Lost or damaged discs were to be replaced so that each group stayed at twelve. None had to be replaced.



3.3 Materials

The materials used were:

1. G-ænial Universal Injectable, shade A2, a highly filled injectable composite (GC Corporation, Tokyo, Japan).
2. Beautifil Flow Plus X F00, shade A2, a highly filled injectable giomer, zero-flow (Shofu Inc., Kyoto, Japan).
3. Beautifil II LS, shade A2, a conventional nanohybrid giomer (Shofu Inc., Kyoto, Japan).
4. Colgate Total toothpaste (relative dentine abrasivity 70).
5. Distilled water.

The approved grouping was:

Group I. G-ænial Universal Injectable.

Group II. Beautifil Flow Plus X F00.

Group III. Beautifil II LS.

The composition of the three resin composites, as listed in the approved protocol, is given in Table 3.1.

Table 3.1 Brand, type, matrix, filler and load of the resin composites used in the study

Composite	Type	Matrix and filler	Load
G-ænial Universal Injectable (A2)	Nanofilled injectable	Dimethacrylate monomers; barium glass and silica	50 vol%
Beautifil Flow Plus X F00 (A2)	Giomer injectable, F00	Bis-GMA, TEGDMA; multifunctional glass and S-PRG	47 vol%
Beautifil II LS (A2)	Nanohybrid giomer	Bis-GMA, UDMA, Bis-EMA, PEGMA, TEGDMA; multifunctional glass and S-PRG	68 vol%

Manufacturers: GC Corporation, Tokyo, Japan; Shofu Inc., Kyoto, Japan. Filler loads are those recorded in the approved protocol (volume percent).



3.4 Equipment

1. Custom-made toothbrushing simulator, Dental Biomaterial Department, Faculty of Dentistry, Alexandria University.
2. Blender computer-aided design software (Blender Institute B.V., Amsterdam, the Netherlands), used to mill the Teflon mould.
3. MarSurf PS10 contact profilometer (Mahr GmbH, Göttingen, Germany).
4. RADWAG AS 220-R2 analytical balance (RADWAG Wagi Elektroniczne, Warsaw, Poland).
5. LED curing light, MiniS (Woodpecker, Guilin, China), 800 mW/cm².
6. Digital caliper.

3.5 Specimen Preparation

Thirty-six discs were packed, one increment each, in a CAD/CAM-milled Teflon mould, 10 mm in diameter and 1 mm deep. A Mylar strip and a transparent microscope slide were placed over the paste so that the face was flat and free of air. The light tip was held against the slide. Each disc was cured for 20 s at 800 mW/cm². Three further 20 s exposures were given from different directions so that the disc was polymerised through its thickness. Flash was trimmed with a no. 12 scalpel blade (Suzhou Kyuan Medical Co., Jiangsu, China). No graded polish sequence was used. The test face is the as-cured, strip-finished surface.

After removal from the mould the diameter of each disc was checked with the digital caliper. The discs were stored in distilled water in a closed glass container at 37 °C for 24 h. That interval is the usual short water soak before a laboratory mechanical test (24,25). Before each weighing or roughness run a disc was taken out and dried with a



sterile disposable sponge. The same drying habit was used at baseline and after brushing so that retained water would not be read as mass.

3.6 Baseline Measurements

Each dried disc was weighed on the RADWAG AS 220-R2. The recorded masses run to 0.0001 g; the balance therefore read to 0.1 mg. Gravimetric loss is the wear outcome named in the protocol (3,15).

Ra was measured on the test face with the MarSurf PS10. Five traces were taken at different sites on each disc. The value used for analysis was the mean of those five traces. Ra is the profile parameter of ISO 4287 (16). Five traces reduce the chance that a single scratch sets the disc value. They do not produce an areal map (23,31).

3.7 Toothbrushing Protocol

Discs were mounted in the custom brushing machine. The vertical load was 2.0 N, inside the ISO range of 0.5–2.5 N (3). The slurry was 250 g of Colgate Total in 1 L of distilled water, after ISO/TR 14569-1 and ISO 11609 (3,14). Relative dentine abrasivity of that paste is 70. The slurry was replaced every 5 000 cycles. Each disc received 10 000 cycles. That count is the conventional laboratory equivalent of one year of twice-daily brushing (7,8). After the run each disc was washed with a stream of tap water and dried with a disposable sponge.

3.8 Post-test Evaluation

Mass and Ra were recorded again with the same instruments, the same drying habit and, for Ra, five traces a disc.

Percentage weight loss was calculated as:



$$\text{Percentage weight loss} = [(W_1 - W_2) / W_1] \times 100$$

where W_1 and W_2 are mass before and after brushing.

$$\text{Absolute loss (mg)} = (W_1 - W_2) \times 1000.$$

$$\Delta Ra (\mu m) = Ra \text{ after} - Ra \text{ before}.$$

3.9 Statistical Analysis

Descriptive values are presented as mean $\pm$ standard deviation. Percentage weight loss was compared among groups by one-way analysis of variance (ANOVA). Ra after brushing was compared by the Kruskal–Wallis test, followed by pairwise post-hoc comparisons. The level of significance was set at $\alpha = 0.05$.

Percentage loss was treated as a parametric outcome. Ra after brushing was not. The locked analysis used those two tests. They are not interchangeable. A later check of normality on the raw sheet would belong in an appendix if that sheet is released. It is not invented here.

3.10 Ethical Considerations

The work used commercial restorative pastes and laboratory discs. No patient, extracted tooth or identifiable human tissue was used. Ethical approval for human subjects was not required.



CHAPTER 4

RESULTS

Thirty-six discs were tested, twelve in each group. All discs completed 10 000 cycles at 2 N. None was lost in processing. Wear is reported as mass, as percentage of the baseline mass and as milligrams shed. Roughness is reported as Ra, as the mean of five traces a disc, before and after the same abrasion.

The three pastes did not start at the same mass or at the same Ra. Those baseline differences are part of the result. They are not adjusted away. All thirty-six faces were strip-finished in the same mould, so the baseline Ra gap is a material difference, not a difference in grit (28,35).

4.1 Weight Loss

Table 4.1 Mean weight before and after simulated toothbrushing and percentage weight loss of the tested composite resins (n = 12)

Composite resin	Weight before (g) Mean $\pm$ SD	Weight after (g) Mean $\pm$ SD	Weight loss (%) Mean $\pm$ SD	p-value
Beautifil Flow Plus X F00	0.1602 $\pm$ 0.0074	0.1576 $\pm$ 0.0031	1.49 $\pm$ 3.14	
G-aenial Universal Injectable	0.1630 $\pm$ 0.0102	0.1619 $\pm$ 0.0106	0.65 $\pm$ 0.37	0.329†
Beautifil II LS	0.2313 $\pm$ 0.0154	0.2303 $\pm$ 0.0156	0.41 $\pm$ 0.25	

† One-way ANOVA. Values are mean $\pm$ standard deviation.

Beautifil II LS discs were heavier at baseline (0.2313 $\pm$ 0.0154 g) than the two injectables (0.1602 $\pm$ 0.0074 g and 0.1630 $\pm$ 0.0102 g). The mould was the same. The difference is filler load and density. Beautifil II LS is listed at 68 vol%. The injectables are

listed at 47 vol% and 50 vol%. A heavier disc of the same diameter and thickness is the expected finding, not a fault in allocation.

One-way ANOVA on percentage loss gave $p = 0.329$. The first null hypothesis is not rejected. Numerically, Beautifil II LS lost the least (0.41 ± 0.25 %), then G-ænial Universal Injectable (0.65 ± 0.37 %), then Beautifil Flow Plus X F00 (1.49 ± 3.14 %). The last standard deviation is an order larger than the other two.

A non-significant p-value does not make the three means equal. It says that, with twelve discs a group and with the variance that was observed, the test could not separate them. Most of that variance sits in one group. Figure 4.1 plots the same means and standard deviations as Table 4.1.

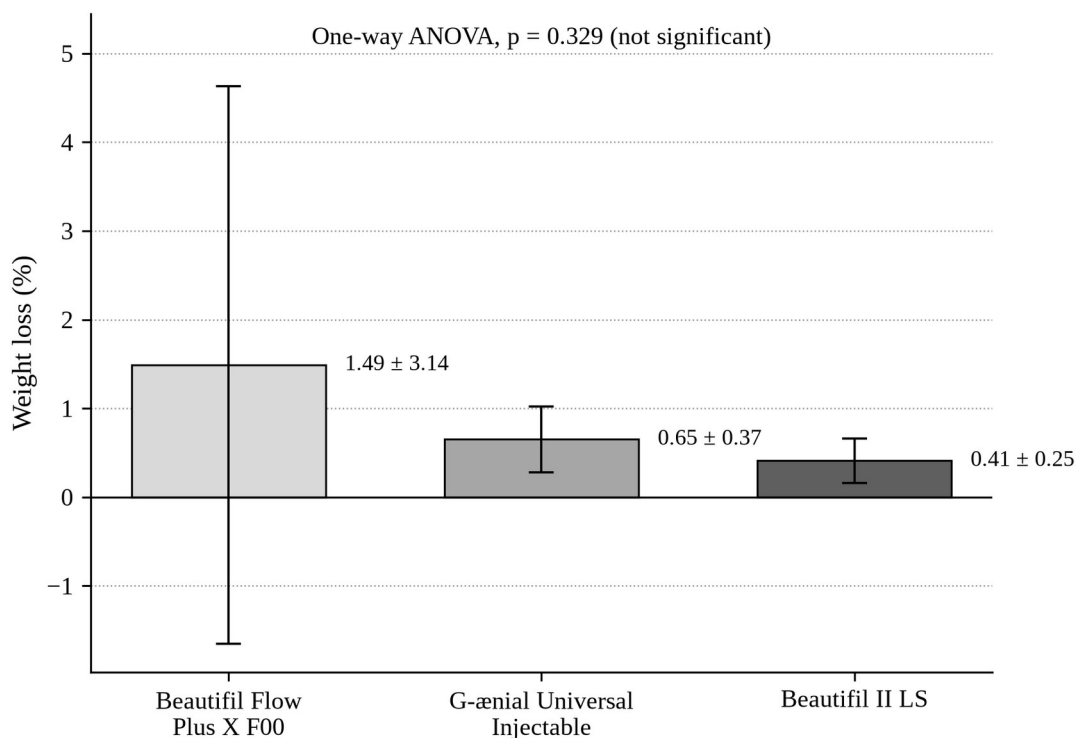


Figure 4.1 Mean percentage weight loss of the three composite resins after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation ($n = 12$ per group). In the Beautifil Flow Plus X F00 group the standard deviation exceeds the mean, so the

lower error bar extends below zero; this reflects the dispersion introduced by a single specimen that lost about 11.4 % of its mass and does not imply a gain in weight.

Differences among groups were not statistically significant (one-way ANOVA, $p = 0.329$).

Absolute loss is given in Table 4.3, after the roughness data, because percentage loss on a lighter disc can exaggerate a small mass change. The two numbers should be read together.

4.2 Surface Roughness

Mean Ra before and after brushing, and ΔRa , are given in Table 4.2.

Table 4.2 Mean surface roughness (Ra) before and after simulated toothbrushing and the change in roughness (ΔRa) of the tested composite resins ($n = 12$)

Composite resin	Ra before (μm) Mean $\pm$ SD	Ra after (μm) Mean $\pm$ SD	ΔRa (μm) Mean $\pm$ SD	p-value
Beautifil Flow Plus X F00	0.038 ± 0.011	0.104 ± 0.017	0.066 ± 0.021	
G-aenial Universal Injectable	0.054 ± 0.010	0.100 ± 0.023	0.046 ± 0.023	$< 0.001\ddagger$
Beautifil II LS	0.119 ± 0.054	0.194 ± 0.050	0.075 ± 0.066	

$\ddagger$ Kruskal–Wallis test. Pairwise tests: each injectable versus Beautifil II LS, $p < 0.001$; the two injectables, not significant. Values are mean $\pm$ standard deviation.

At baseline Beautifil II LS was already the roughest ($0.119 \pm 0.054 \mu m$). Beautifil Flow Plus X F00 was the smoothest ($0.038 \pm 0.011 \mu m$). G-aenial Universal Injectable lay between them ($0.054 \pm 0.010 \mu m$). The finishing step was the same for every disc: a Mylar strip under a glass slide and a scalpel for flash. No further polish sequence was used. The baseline gap is therefore a material difference on an as-cured, strip-finished face, not a difference in grit.

Every group roughened. ΔRa was $0.066 \pm 0.021 \mu m$ for Beautifil Flow Plus X F00, $0.046 \pm 0.023 \mu m$ for G-aenial Universal Injectable and $0.075 \pm 0.066 \mu m$ for Beautifil II LS. After 10 000 cycles the means were $0.104 \pm 0.017 \mu m$, $0.100 \pm 0.023 \mu m$ and $0.194 \pm 0.050 \mu m$. Figure 4.2 shows Ra before and after on the same discs. Figure 4.3 shows ΔRa .

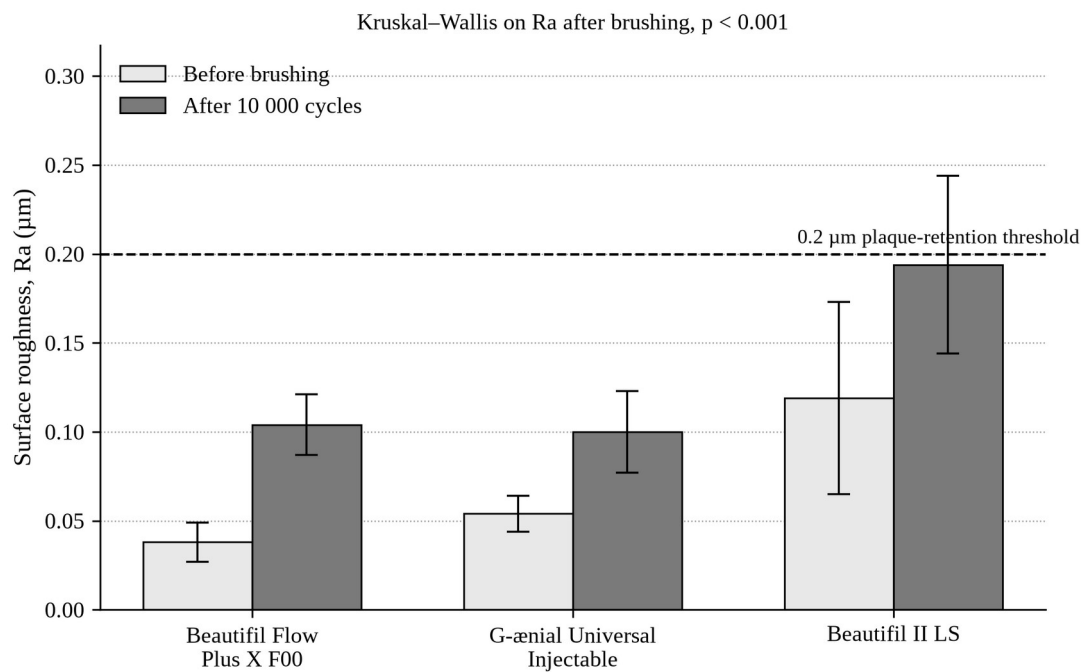


Figure 4.2 Mean surface roughness (Ra) of the three composite resins before and after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation ($n = 12$ per group). The broken line marks the $0.2 \mu m$ threshold above which plaque retention increases. Ra after brushing differed significantly among groups (Kruskal–Wallis, $p < 0.001$).

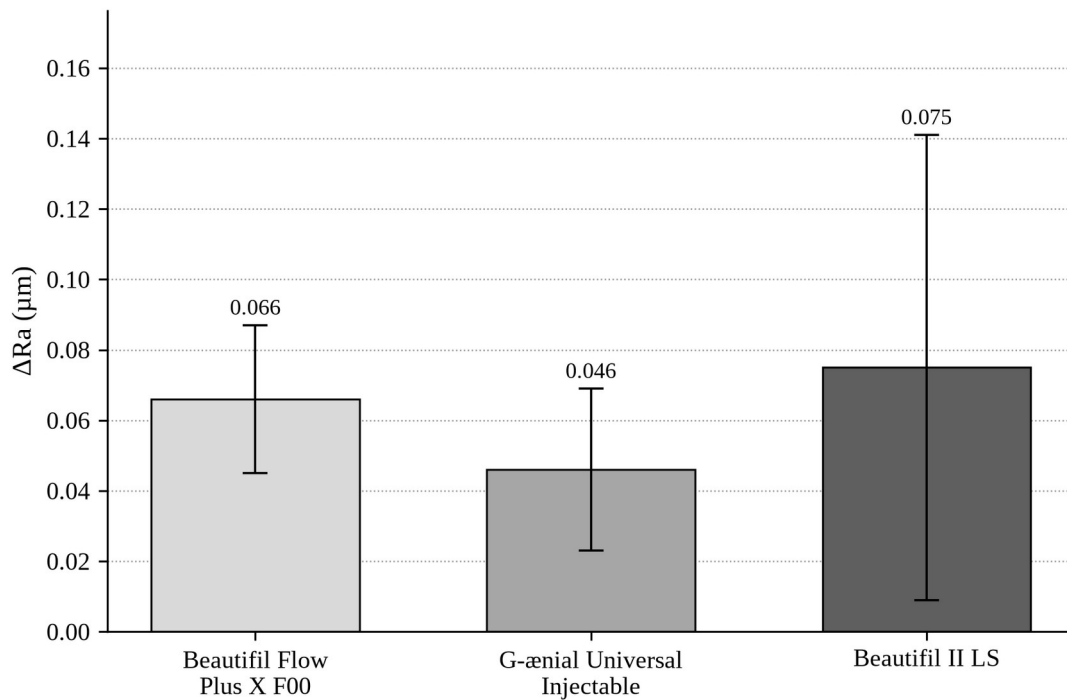


Figure 4.3 Mean change in surface roughness ($\Delta Ra = Ra$ after – Ra before) of the three composite resins after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation ($n = 12$ per group).

From the same locked means, Ra rose by a factor of about 2.7 in Beautifil Flow Plus X F00 (0.038 μm to 0.104 μm), about 1.9 in G-aenial Universal Injectable (0.054 μm to 0.100 μm) and about 1.6 in Beautifil II LS (0.119 μm to 0.194 μm). The nanohybrid started higher, so its relative rise is smaller even though its absolute ΔRa is the largest.

The Kruskal–Wallis test on Ra after brushing was significant ($p < 0.001$). The second null hypothesis is rejected. Pairwise tests separated Beautifil II LS from each injectable. They did not separate the two injectables.

The order of the groups did not change. The nanohybrid was the roughest before and the roughest after. G-aenial Universal Injectable changed the least. Beautifil Flow Plus X F00 started lowest and finished with G-aenial Universal Injectable.

Relative to the 0.2 μm plaque line (1), both injectable means were below it. The Beautifil II LS mean was 0.194 μm , on the line. Its standard deviation of 0.050 μm means some discs in that group crossed 0.2 μm . All three means were well below 0.50 μm (2). A patient would not be expected to feel these surfaces after this brushing model. Plaque retention is the sharper distinction, and it applies to the nanohybrid group.

Absolute weight loss is set out in Table 4.3.

Table 4.3 Absolute weight loss after 10 000 brushing cycles (n = 12)

Composite resin	Absolute weight loss (mg) Mean $\pm$ SD	Range (mg)
Beautifil Flow Plus X F00	2.58 $\pm$ 5.91	0.2 – 20.7
G-aenial Universal Injectable	1.07 $\pm$ 0.62	0.2 – 1.9
Beautifil II LS	0.95 $\pm$ 0.58	0.1 – 1.8

Values are mean $\pm$ standard deviation. The large standard deviation in Beautifil Flow Plus X F00 comes from one disc that lost about 11.4 % of its mass (20.7 mg). The rest of that group, and both other groups, sit in a band of roughly 0.1–1.9 mg.

4.3 Summary of Results

Percentage weight loss did not differ significantly ($p = 0.329$). Absolute loss in the two well-behaved groups and in most of the Beautifil Flow Plus X F00 discs was about 1 mg. One disc in that group lost 20.7 mg. After the same abrasion both injectables were smoother than Beautifil II LS ($p < 0.001$) and did not differ from each other. The injectable means stayed below 0.2 μm . The nanohybrid mean sat on that line.



CHAPTER 5

DISCUSSION

The first null hypothesis stands. The second does not. The three resins lost a statistically similar fraction of their mass. They did not finish with a similar Ra.

The protocol asked whether two current injectables would wear and roughen like Beautifil II LS under toothbrush slurry. The answer is split. On mass they could not be separated. On Ra they could.

5.1 Weight Loss

A non-significant ANOVA ($p = 0.329$) is not proof of equal wear. It is a failure to reject equality. Beautifil Flow Plus X F00 had a standard deviation of 3.14 % beside a mean of 1.49 %. That spread is the outlier disc (about 11.4 %, 20.7 mg). Residual variance rises. Power falls. The other eleven discs in that group, and the twenty-four discs in the other groups, lost between 0.1 mg and 1.9 mg. On that restricted view the three pastes are close.

Beautifil II LS started heavier, as its 68 vol% listing would lead one to expect. Percentage loss on a heavier disc is a smaller number for the same milligrams shed. Absolute loss treats that fairly: 0.95 ± 0.58 mg against 1.07 ± 0.62 mg for G-ænial Universal Injectable. Those two means are near each other. The injectable with the barium-glass and silica bed (50 vol%) did not shed more mass than the conventional nanohybrid under this slurry.

Condon and Ferracane placed a wear step at 48 vol% in an oral-wear simulator (12). The protocol lists Beautifil Flow Plus X F00 at 47 vol% and G-ænial Universal



Injectable at 50 vol%. Those two sit on either side of that step. Beautifil II LS sits well above it. The present mass result did not follow that step. The test is toothbrush slurry, not their simulator, and the filler chemistry is not their experimental series. The volume figures are still the right frame for why a 68 vol% paste was expected to lose less, and why that expectation was not confirmed on percentage loss.

The slurry was a single paste, Colgate Total, RDA 70, mixed 250 g to 1 L after ISO/TR 14569-1 (3,14). That is a moderate abrasive, not a high-RDA whitening paste. Monteiro and Spohr showed that a higher RDA cuts gloss faster and leaves a rougher track (8). A harsher dentifrice might have opened a gap in mass that this paste did not. The present ranking is for this slurry only.

Rajabi et al. found less volumetric two-body wear for G-aenial Universal Injectable and Beautifil Flow Plus F00 than for a nanohybrid paste (6). The direction is friendly to injectables. The test is not the same test. They used 200 000 chewing cycles, 49 N and a steatite ball. They did not use toothpaste. They measured volume. Beautifil Flow Plus F00 is also not Beautifil Flow Plus X F00. Agreement can be stated only as a general remark: highly filled injectables are not automatically the weaker partner in a wear assay.

Why one disc lost 20.7 mg is not known from the laboratory record used here. A void, a soft under-cured layer, or a weighing error would all produce that point. Curing was 20 s through the slide plus three further 20 s exposures. That is a full dose for a 1 mm disc. It does not exclude a local defect in one increment. The value was not dropped. It is footnoted in Table 4.3. Any later re-analysis that excludes it must say so.

Gravimetric loss is a bulk number. It does not say where the substance left the disc. Heintze compared mass, profile depth and scanned volume and showed that the three

methods do not tell the same story (15). The present work has only mass. A later volumetric or profilometric wear depth would sit beside Table 4.3, not replace it.

5.2 Surface Roughness

The roughness result is cleaner. The nanohybrid was rougher before brushing and rougher after it. ΔRa was also largest in that group, with a wide spread ($0.075 \pm 0.066 \mu m$). G-aenial Universal Injectable changed the least ($0.046 \pm 0.023 \mu m$). Beautifil Flow Plus X F00 started smoothest and ended statistically with G-aenial Universal Injectable.

That pattern fits filler geometry. A nanofilled injectable with barium glass and silica at 50 vol% wears more evenly (6,18,28). A nanohybrid with larger S-PRG and glass particles loses matrix first. The large fillers stand up or leave pits (4,18,37). Cavalcante et al. showed the same size effect on model pastes after 20 000 strokes (28). Senawongse and Pongprueksa, and Suzuki and Finger's group, saw a smoother nanofill than nanohybrid after brushing (35,37). Hasan, Eltoukhy and Zaghloul saw the same order with G-aenial Universal Injectable against a nanohybrid (18). Heintze et al. had already shown that hybrid and microhybrid pastes roughen more with load and time than microfilled resins (4).

The slurry here was RDA 70. da Costa and Monteiro both showed that a harsher dentifrice raises Ra further on the same composite (8,29). The present ranking is for this paste.

The discs were strip-finished, not polished through a graded grit sequence. Baseline Ra is therefore the as-cured skin plus whatever the scalpel left at the edge. The after-brushing comparison is still valid. Every group had the same finish and the same 10 000



cycles. Part of the after-brushing gap, though, was already present at baseline. Beautifil II LS started at $0.119\ \mu\text{m}$. The injectables started at $0.038\ \mu\text{m}$ and $0.054\ \mu\text{m}$.

Five traces a disc on the MarSurf PS10 is a standard contact-stylus habit after ISO 4287 (16). It is not a map of the whole face. A local pit can be missed. Optical methods, or AFM, would have given that map (23,31). They were not used. Kakaboura showed that rank order usually survives a change of method even when the micrometre values do not (31).

The $0.2\ \mu\text{m}$ line is the clinically useful one (1,17). Both injectables ended near $0.10\ \mu\text{m}$. They remain, on the mean, in the range where further smoothing is not expected to cut plaque. Beautifil II LS ended at $0.194 \pm 0.050\ \mu\text{m}$. The mean is acceptable on the published threshold. The spread is not entirely below it. None of the means approach the $0.50\ \mu\text{m}$ tongue threshold (2). Patients would not be expected to feel these surfaces after one year of this brushing model. Biofilm risk is the sharper distinction, and it applies to the nanohybrid group.

S-PRG glass can roughen in acid by ion exchange (21). The present slurry was not an acid challenge. The higher Ra of Beautifil II LS after brushing is read as filler geometry under three-body abrasion, not as a fluoride-release effect. An eight-year clinical series of an earlier giomer paste showed acceptable retention (22). That is a different question from toothbrush Ra of Beautifil II LS.

5.3 Clinical Implications

Within this protocol the two injectables may be used where toothbrush abrasion is the wear of interest without paying a roughness penalty against Beautifil II LS. They do not, on these data, wear less. They finish smoother. That matters on free cleanable surfaces



and at margins of anterior restorations, which is the wear mode the protocol named. It does not license a claim that they survive occlusal contact better. Occlusal contact was not tested.

Beautifil II LS remains a reasonable packable control. Its higher baseline Ra is the as-cured nanohybrid face. If a clinic polishes that paste through a fine sequence, the starting Ra would be lower than 0.119 μm . The after-brushing test would then start from a different place. The groups here were compared as they stood after the same strip finish and the same abrasion.

The injectable means at 0.10 μm sit in the range where Bollen and Quirynen would not expect a further plaque gain from smoothness (1,17). The nanohybrid mean at 0.194 μm is on their line. Some discs in that group are over it. That is the clinical sentence the roughness data will bear. It will not bear a sentence about occlusal longevity or about stain, which was not scored.

5.4 Strengths of the Study

The three products are current commercial pastes, including the X-generation Flow Plus, not the older Beautifil Flow Plus F00. Sample size was twelve, not the ten given by the first G*Power run. Both mass and Ra were read on the same discs. The mould, the cure, the balance, the profilometer, the load, the slurry ratio and the cycle count are those of the approved protocol. The brushing load and cycle count sit on published ISO and Sexson–Phillips figures (3,7). The statistical tests match the distributions used in the locked spreadsheet. The outlier was kept and named.



5.5 Limitations of the Study

The test is in vitro. Saliva, pH, enzymes and occlusal load are missing. Gravimetric loss is not worn volume and is not worn anatomy (15). One dentifrice was used. There are no scanning electron images, so filler plucking is an inference from the roughness numbers, not a micrograph. Ten thousand cycles are one conventional year, not a clinical year (7,8). Disc geometry is not a Class II box. The face was strip-finished, not polished. Storage was 24 h in water at 37 °C; longer water ageing was not done.

5.6 Concluding Remarks

After 10 000 strokes at 2 N the three resins could not be separated on percentage weight loss. They could be separated on Ra. The injectables were smoother. Beautifil II LS sat on the plaque threshold. That is the result. It should not be enlarged into a general ranking of injectable versus nanohybrid outside toothbrush abrasion.



CHAPTER 6

CONCLUSIONS AND RECOMMENDATIONS

6.1 Summary

Thirty-six discs, 10 mm × 1 mm, of Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS were brushed for 10 000 cycles at 2 N in a Colgate Total slurry (RDA 70; 250 g/L). Mass was read on a RADWAG AS 220-R2. Ra was the mean of five MarSurf PS10 traces. Percentage weight loss did not differ (ANOVA $p = 0.329$). Ra after brushing did (Kruskal–Wallis $p < 0.001$). The injectables were smoother than Beautifil II LS and not different from each other. One Flow Plus X disc lost about 11.4 % of its mass.

6.2 Conclusions

Under the conditions of this in-vitro study:

1. Percentage weight loss after 10 000 toothbrushing cycles did not differ significantly among Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS ($p = 0.329$).
2. Ra after the same abrasion did differ ($p < 0.001$). Both injectable resins were smoother than Beautifil II LS. The two injectables did not differ from each other.
3. Mean Ra of both injectables remained below 0.2 μm . Mean Ra of Beautifil II LS was 0.194 μm .
4. One Beautifil Flow Plus X F00 disc lost about 11.4 % of its mass and is the source of that group's large standard deviation.
5. The first null hypothesis is retained. The second is rejected.



6.3 Recommendations

Clinical practice. Where toothbrush abrasion and smoothness after hygiene are the concern, either injectable tested here may be considered beside Beautifil II LS. The data do not show less wear. They show a smoother surface after this brushing model.

Further laboratory work. Repeat the mass analysis with and without the outlying disc, and report both. Add volumetric loss or profilometric wear depth. Add SEM of the worn face. State the polish sequence, the brush and the slurry ratio in full. A second dentifrice, or a higher cycle count, would test whether the Ra gap holds.

Clinical research. Only a controlled clinical study can say whether the 0.10 μm versus 0.19 μm difference after a laboratory year changes plaque, stain or survival.



CHAPTER 7

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الملخص العربي

أُجريت هذه الدراسة في المختبر لمقارنة مقاومة التآكل وخشونة السطح لراتينجين مركبين قابلين للحقن عاليي الملء (Beautifil Flow Plus X F00 و G-ænial Universal Injectable) مع راتينج نانوهجين تقليدي (Beautifil II LS) بعد عشرة آلاف دورة تفريش بحمولة 2 نيوتن في معلق معجون Colgate Total. أُعدت الأقراص بقطر 10 مم وسمك 1 مم (اثنا عشر قرصاً لكل مجموعة) في قالب تفلون، وحُزنت في ماء مقطر عند 37 مئوية مدة أربع وعشرين ساعة. قيست الكتلة بميزان RADWAG AS 220-R2، وقيس Ra بمتوسط خمس قراءات على جهاز MarSurf PS10. لم يبلغ الفرق في النسبة المئوية لفقد الوزن حد الدلالة الإحصائية (تحليل التباين الأحادي، $p = 0.329$). أما Ra بعد التفريش فقد اختلف اختلافاً دالاً (كروسكال-واليس، $p < 0.001$). كان القابلان للحقن أنعم من Beautifil II LS ولم يختلفا أحدهما عن الآخر. بقي متوسط القابلين للحقن دون 0.2 ميكرومتر، ووقع متوسط Beautifil II LS على 0.194 ميكرومتر. فقد قرص واحد من Beautifil Flow Plus X F00 نحو 11.4% من كتلته ولم يُستبعد من التحليل. لا تجيز هذه النتائج الادعاء بأن الراتينجات القابلة للحقن أفضل ترميمات تحت الحمل الإطباق. الاختبار هو سحل فرشاة الأسنان ثلاثي الأجسام، وليس السحل ثنائي الأجسام بخصم مضاد.